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UltraVI80 Instrument Board : Features

Features

The UltraVI80 instrument board includes the following features:

- 80 DCVI channels at -2 to +7V, 1A continuous
  - 2A on a limited simultaneous number of channels, with a restricted voltage range of -1V to 6V
  - Merge on the DIB to achieve up to 4A
  - Ground referenced, 16 bit resolution
- Four quadrant programmable V/I capability
- 8 DC Differential Voltmeters (DCDiffMeters)
- 4 DC Time Measurement Units (DCTimes)
- Per channel access for DC Time Measurement Units and DC Differential Voltmeters
- Dedicated High Performance access for each DCTime
- AWG and Capture with 512K of memory per pin
- PSet controllable
- Fast data readback through Instrument Initiated Moves (IIM) and DMA
- On-board hardware averaging
- Integrated Force Voltage Focus Calibration
- Voltage Spike Check function monitor access per channel

DCVI
UltraVI80 DCVI includes:

- 80 DCVI channels
- All channels operate in four quadrants, forcing positive or negative voltage, sourcing or sinking current
- Voltage range: -2V to 7V
- Seven current ranges: 20uA, 200uA, 2mA, 20mA, 200mA, 1A, and 2A
- Can be referenced to one of two DGS lines available per group of 20 VI channels
- Force, Sense, and Guard lines (Guard permanently connected)
- Per channel voltage and current meter with 10kHz and 100kHz analog filters
- Per channel AWG and Digitizer
- IIM, DMA, hardware averaging
- Per channel Voltage Spike detection circuitry
- Merge two channels on the DIB to get up to 4A
- Syntax compatible with DC30/75
- Bleeder resistor to help discharge DIB capacitance

DC Differential Voltmeter

The UltraVI80 DC Differential Voltmeter (DCDiffMeter) is similar to the DCDiffMeter on the DC30. The UltraVI80 includes:

- 8 DC Differential Voltmeters (DCDiffMeters)
    - Accessed through VI channel sense lines

- Can connect to one of two DGS lines available per group of 20 VI channels
- Precision voltage measurement capability

To improve accuracy, two separate analog-to-digital converters are included, available in two modes of operation:

- High Accuracy
  - 24-bit oversampled SAR ADC
  - 1K strobes memory
  - Waveform captures not supported
  - Sample rate of 500 sps to 100 ksps, adjustable with 10s resolution
- High Speed
  - 16-bit ADC
  - 1k strobes memory
  - 4M waveform samples memory
  - Sample rate up to 8.34Msps adjustable with 120ns resolution
- All DCDiffMeters have:
  - Measure voltage ranges of 1.4V, 3.5V, and 7V
  - Hardware averaging: 2, 4, 8, 16, 32, 64, 128, and 256 samples can be averaged

- Syntax compatible with DC30/75

DC Time Measurement Unit

The UltraVI80 DC Time Measurement Unit (DCTime) is similar to the DCTime on the DC30. The voltage range is asymmetric, restricted to -2V to 7V. The UltraVI80 includes:

- 4 DC Time Measurement Units
  - Accessed through VI channel sense lines or dedicated high performance inputs
  - Can connect to one of four DGS lines available per group of 40 VI channels
  - Can asynchronously trigger any DCVI Meter and DCDiffMeter on the same group of 40 channels (channels 1-40 or channels 41-80)
- Syntax compatible with DC30/75

DGS

- Eight Device Ground Senses per instrument, two per slice of 20 channels
- Any channel in a slice can be referenced to either of the two DGS dedicated to its slice

POP

- Pattern Oriented Programming through PSets and microcodes
- 4K ICS codes

Arbitrary Waveform Generation and Capture

- AWG per channel
  - 16 bit DAC

- 512k samples memory per channel
- Up to 1Msps sampling rate per channel adjustable in 1s steps (from 1s to 1023s)

- Digitizer per channel
  - 16 bit ADC
  - 512k samples memory per channel for signals, 1k samples for strobes
  - Up to 1Msps sampling rate per channel adjustable in 1s steps (from 1s to 1280s)
  - 10kHz and 100kHz analog filters

Calibration

- Integrated Force Voltage calibration
- Uses DCDiffMeter to perform a calibration on the fly
- Calibration remains valid under the following conditions:
  - The current load remains static for 100ms before the force precision application and throughout a valid force precision window
  - Force precision window is valid for 1 second after force precision application
  - Valid ranges: 20A, 200A, 2mA, 20mA, and 200mA

Data Throughput

- Improved direct memory access (DMA)
- Instrument Initiated Move (IIM) over MoveLink

- Hardware averaging
- Hardware calibration factors

Voltage Spike Check Tool
Per channel Spike Check detection functionality

- Integrated instrument functionality to check for spikes in a device test program
- Reduces the time required to check for spikes using an oscilloscope at the back end of test program qualification
- Uses the per pin voltage clamping detection circuitry, which is a 16-bit DAC with a resolution of 200V and a response time of 1s.

Pattern Instrument Unlock
Pattern unlock is used primarily during debug. It allows you to modify the instrument state and read back measurements from either the debug display or VBT while any pattern is running.
Input Protection from -6V to 30V
The UltraVI80 has input protection from -6V to 30V. This allows you to have an UltraVI80 channel and a DC30 channel connected to the same pin on the DIB without the need for a relay to disconnect the UltraVI80 channel while forcing up to 30V with the DC30.
Sync Panel

- UltraVI80 provides sync panel signal support for the following signals:
  - DCDiffMeter ADC Input (1 signal/DCDiffMeter)
  - DCDiffMeter ADC Start (1 signal/DCDiffMeter)
  - DCTime Async Trigger 1/Trigger 2 (2 signals/DCTime)
  - DCVI Real Time Alarm (1 signal/channel/DCTime)
  - DCVI ADC Start (1 signal/channel)

- DCTime Real Time Shutdown (1 signal/channel)

- UltraVI80 does not support the sync panel signal DCVI ADC Input

Gate Off Hi-Z

- The UltraVI80 provides an alternate Gate_Off_HiZ state.

- When the Gate_Off_HiZ is applied, the following sequence occurs:
  - Set current range to 20A
  - Set force DAC to 0V
  - Set instrument mode to “tlDCVIModeVoltage”

Nominal Bandwidth

IG-XL provides the following interface to read or write the Nominal Bandwidth parameter:

TheHdw.DCVI.Pins(PinList).NominalBandwidth.Value = newVal

Note that the nominal bandwidth is validated against bandwidth settings of 0 to 1003. The default value is 0.

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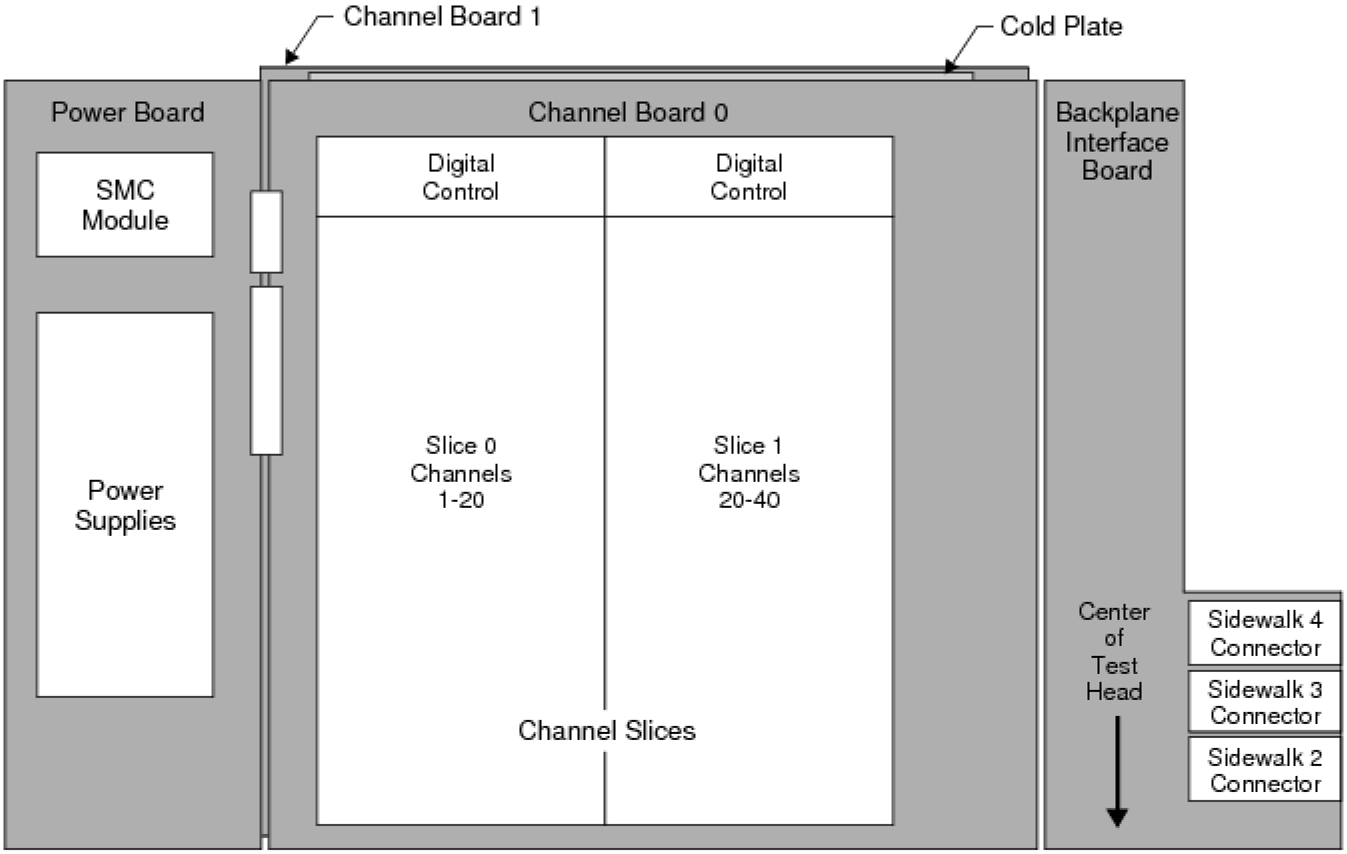
UltraVI80 Instrument Board : UltraVI80 Board Hardware

UltraVI80 Board Hardware

The UltraVI80 instrument board assembly occupies a single slot in the UltraFLEX test head and consists of:

- Two identical 40 channel boards mounted on opposite sides of a liquid-cooled cold plate
- Two channel slices on each channel board that contain:
  - 40 DCVI channels
  - 2 DC Time Measurement Units
  - 4 DC Differential Voltmeters
  - Digital controls
- A backplane interface board (BIB) that provides the connections to the UltraFLEX backplane and manages signal flow between the test system and channel boards
- A power board that converts the +48V system power to the various on-board power voltages that the UltraVI80 board requires

The following figure shows the layout of an UltraVI80 instrument board.



UltraVI80 Instrument Board Hardware Block Diagram

Channel Boards

The channel boards contain the primary DCVI instrument control and tester communication components as well as the output stage of the DCVI channels. The channel boards also have the following features:

- An interface to communicate with the databus of the tester computer
- A TCIO databus connecting to the instrument hardware
- A MoveLink bus for fast transfer of capture data to the Digital Signal Processing (DSP) hardware

Channel Slices

The UltraVI80 has four channel slices, two on each channel board, that contain the analog circuitry of 80 DCVI channels. Each slice supports:

- One group of 20 DCVI channels
- Two DC Time Measurement Units

- Two DC Differential Voltmeters

Channels can be merged to increase the maximum output current and power delivered to the DUT. Four slice digital control FPGAs, two on each channel board, support the module board hardware. Each slice FPGA supports 20 DCVI channels. The DCVI channels’ output stage is also on the main board, attached to the cooling assembly.

Each channel slice of 20 channels also includes the following resources:

- DDR2 memory
- Precision reference and buffering
- Current limiting and bandwidth control hardware
- ADC to meter voltage and current
- On-board calibration and checkers resources
- Parameter set (PSet) memory

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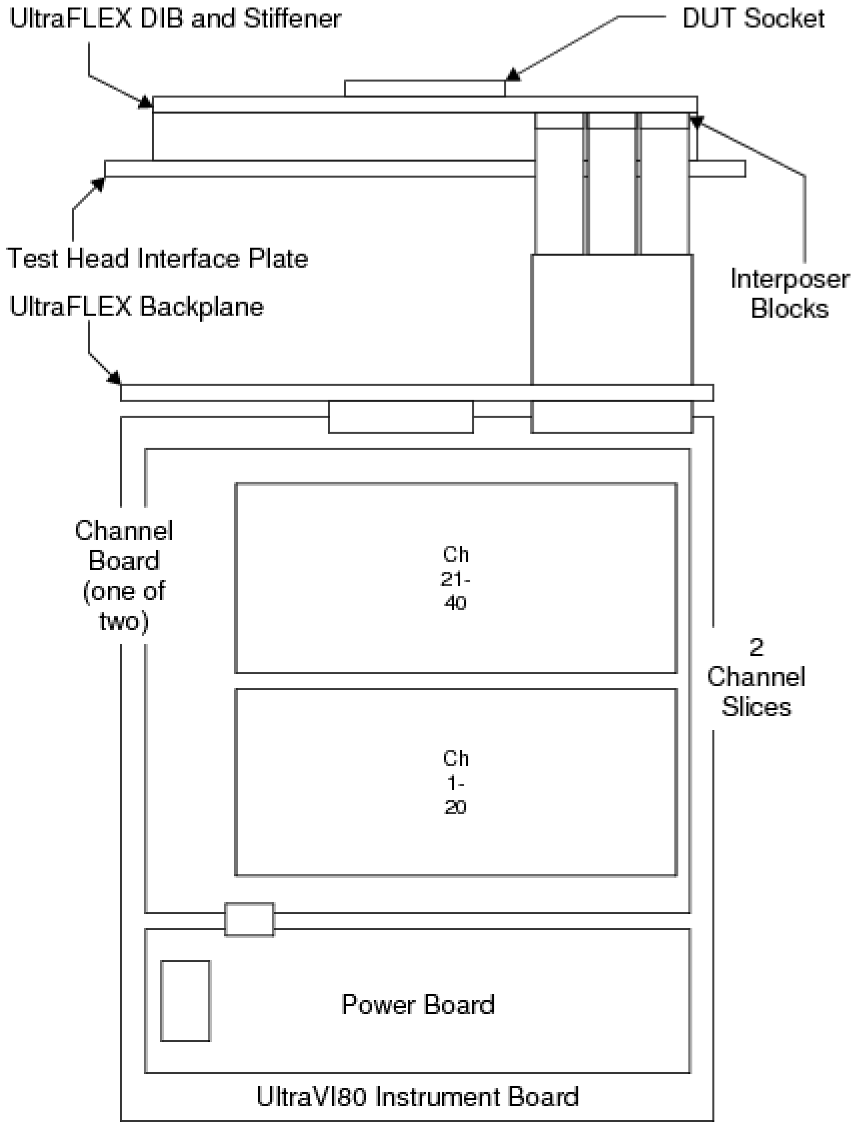
UltraVI80 Instrument Board : Test Head Connections

Test Head Connections

This section shows where the boards are connected to the test head backplane.

The following figure shows how the instrument board is connected through interposer blocks to deliver power to the DUT.



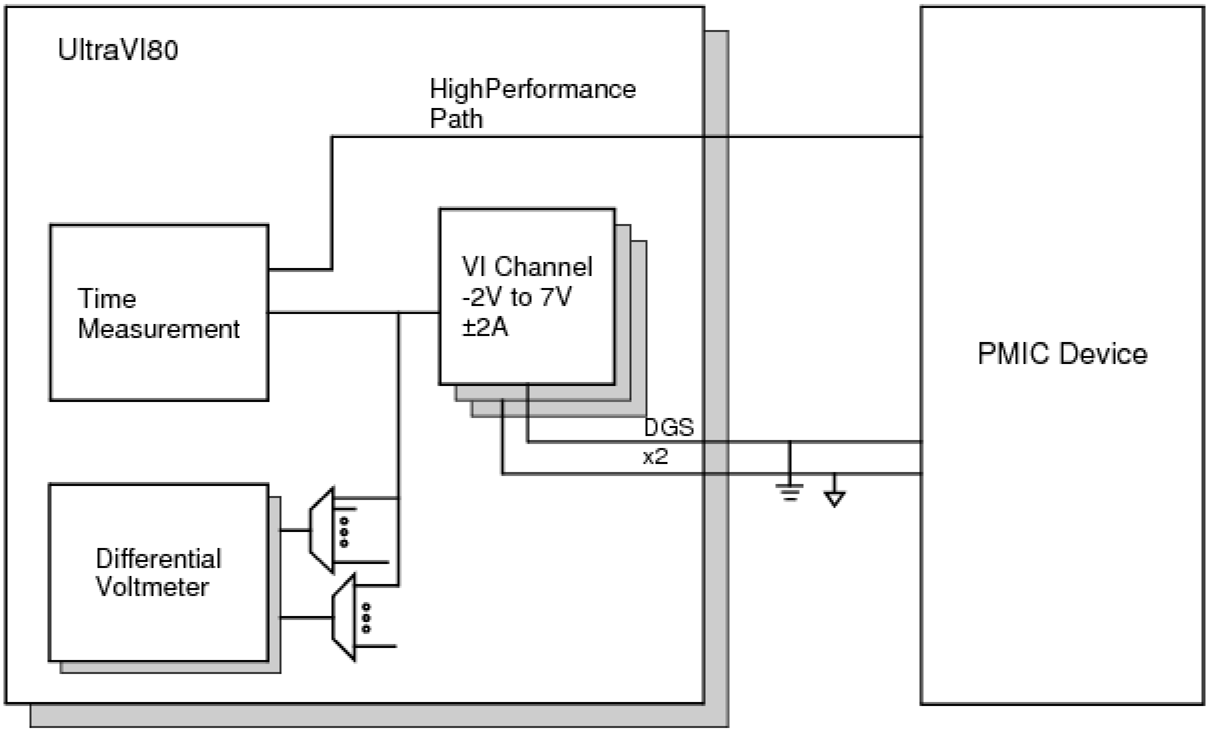
UltraVI80 Instrument Board Connection to the DUT

Several UltraVI80 boards can be installed in an UltraFLEX test head. An UltraVI80 can be installed in any available slot in the test head. Channels are connected to the DIB using three interposer blocks occupying sidewalks 2, 3, and 4 of the loadboard end.

The UltraVI80 test head interface features a high-performance connection to support high current to the DUT through a low impedance path. This interface allows each DCVI channel to respond rapidly to current transients, which helps reduce the need for large amounts of power supply bypass capacitance on the DIB. This saves space on the DIB for additional test sites, further enabling testing of multiple devices at once.

For more information, see UltraVI80 DIB Design Guidelines.

The following figure shows a connection model for the UltraVI80 signals to the DUT.



UltraVI80 Channel Connections to the DUT

Each UltraVI80 channel has an ADC that is used for:

- Force voltage measurement and connected to internal GND or external DGS
- Current measurement

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UltraVI80 Instrument Board : Simulated Configuration Files

Simulated Configuration Files

You can use the UltraVI80 offline by forcing IG-XL to map the UltraVI80 (denoted in configuration files as “DC07”). Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the UltraVI80 (DC07) board in slot 16:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type              idprom (type serial rev company)

4.0            HSD-M            979-219-00 0000000 0000-A 0000
               HSD-M            956-361-00 0000000 0000-A 0000

12.0           HexVS            974-205-00 0000000 0000-A 0000
               HexVS            939-265-00 0000000 0000-A 0000
```

```
16.0        DC-07            604-375-00 00000000 0000-A 5445
            DC-07            939-420-00 00000000 0000-A 5445
            DC-07Power       608-807-00 00000000 0000-A 5445
            DC-07Rider-0     604-534-00 00000000 0000-A 5445
            DC-07Rider-1     610-617-00 00000000 0000-A 5445

21.0        VSM              974-296-00 0000000 0000-A 0000
            VSM              956-361-00 0000000 0000-A 0000

24.0        SupportBoard     974-220-12 0000000 0000-A 0000
            SupportBoard     956-354-00 0000000 0000-A 0000

62.0        DistributionBoard  956-355-00 0000000 0000-A 0000
            DistributionBoard  956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>
```

To use the UltraVI80 offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the UltraVI80 online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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UltraVI80 Instrument Board : UltraVI80 Channel Connections

UltraVI80 Channel Connections

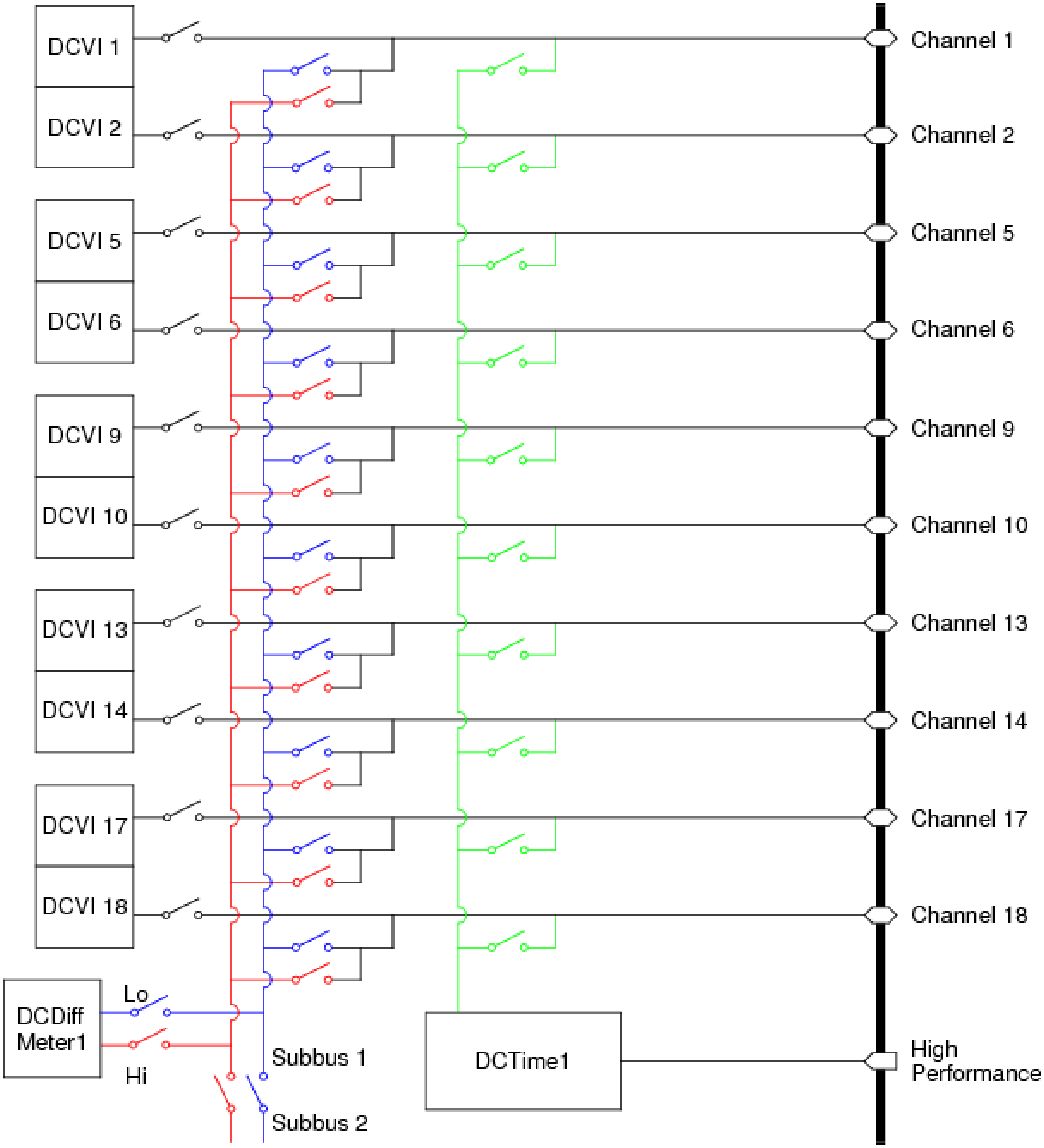
Each UltraVI80 slice has 20 DCVI channels, 2 DC Differential Voltmeters, and 2 DC Time Measurement Units that share connections to the DUT.

Each DC Differential Voltmeter in a slice is located on one of two subbusses allowing direct connections between 10 channels and indirect connections to 10 channels.

The 2 DC Time Measurement Units are shared across the 2 slices on a channel board, with a designated set of 10 DCVI channels per DCTime per slice. This figure shows the connections shared between 10 DCVI channels, one DC Differential Voltmeter, and one DC Time Measurement Unit. Connections for the channel are simplified showing the force, sense, and guard as one line. Plan DIB layout carefully.

☑ **Note:**

Avoid connecting a DCVI to a live (powered) DUT pin. Doing so may potentially damage the DUT or adversely affect instrument reliability mainly due to the relays being hot switched.

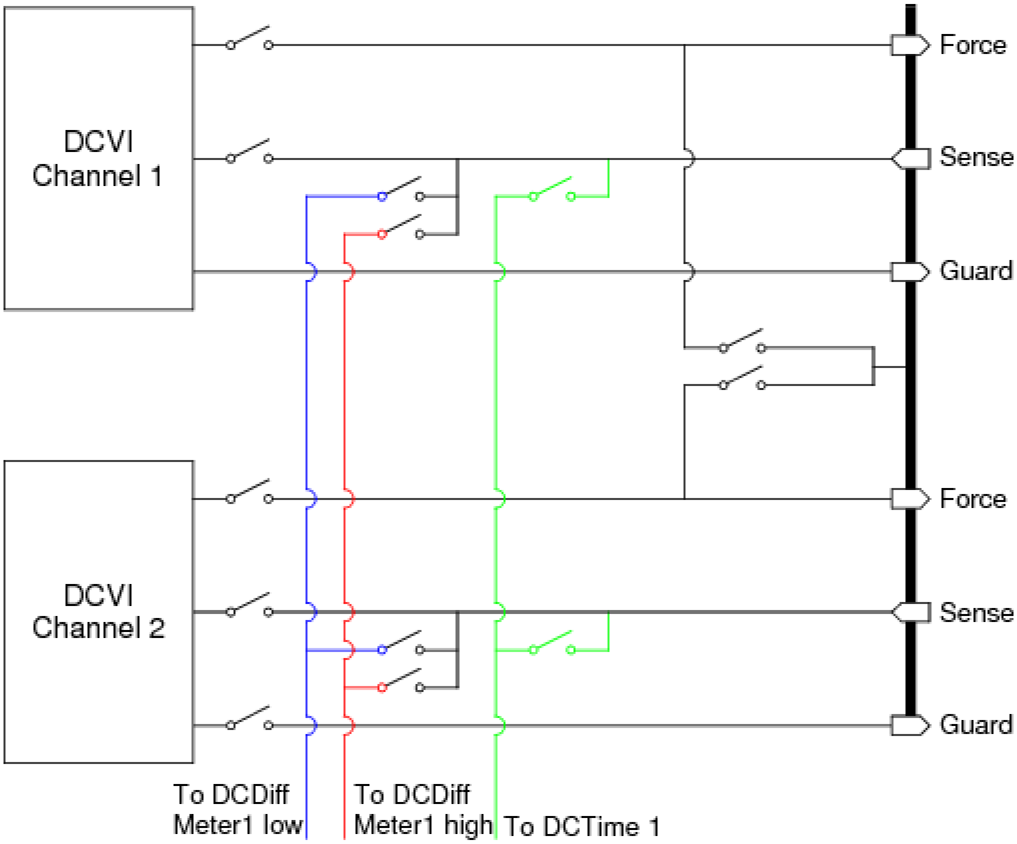


Connections Diagram for 10 UltraVI80 Channels

You can make the following connections:

- DCVI to the DIB through force and sense lines
- DCDiffMeter from the DIB through sense lines
- DCTime from the DIB through the high performance lines
- DCTime from the DIB through sense lines

The following figure shows the force, sense, and guard lines independently to show the connections between the three on-board instruments in greater detail.



Detailed Connections for Two UltraVI80 Channels

For more information, see:

- UltraVI80 DGS Lines
- Connecting a DCDiffMeter
- Connecting a DCTime
- Connecting a DCVI

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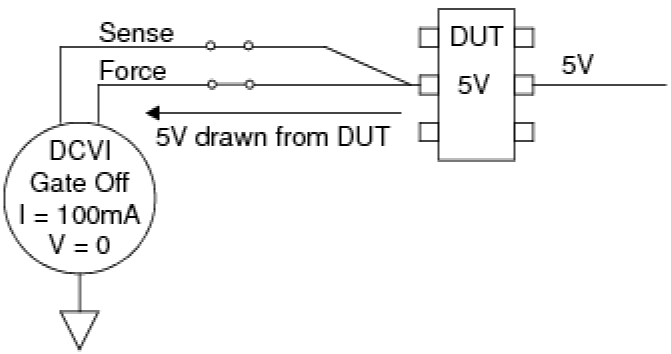
UltraVI80 Instrument Board : Connecting to a Live DUT from Force (V or I) Mode

Connecting to a Live DUT from Force (V or I) Mode

If you must connect to a live DUT from Force (V or I) Mode, use the following best practices.

Potential problems of connecting to a live DUT are shown in the figure.



Connecting to a Live DUT Pin

The following conditions exist in the setup shown in this figure:

- Gate is Off
- Voltage = 0V (Default Gate Off setting)
- Current = 100mA
- DUT is at a 5V potential

Before connecting a DCVI to the DUT, a 5V potential exists between the DCVI and the DUT. As the DCVI relay closes, current will flow between the DCVI and the DUT. This current can damage the relay.

Take the following steps before connecting to minimize the impact of hot switching DCVI relays:

- Program the current range to the lowest instrument range.
- Program the DCVI to the slowest bandwidth setting.
- Program the DCVI voltage as close as possible to the DUT pin.
- Program the Gate to On.
- Program a wait settling time of about 2ms.
- Connect the DCVI.
- Wait the relay settling time.
- Reprogram the voltage and current values required for the test.

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UltraVI80 Instrument Board : Device Ground Sense

Device Ground Sense

Device Ground Sense (DGS) is a sense signal that tracks the voltage of ground on the DUT and is used as a correction by the instrument to accurately force or measure voltages relative to the ground potential of the device. DGS must be connected to a single point on the DIB. The optimum place to connect DGS is at the device ground pin.

DGS connections are made by IG-XL when the instrument is connected to the DUT. Any used DGS must be grounded on the DIB or a DGS alarm can occur.

UltraVI80 DGS Lines

The UltraVI80 Instrument board has eight DGS lines (two per 20 channel slice) that each DCVI and DCTime connect to. You can specify DGS connections in a channel map. Otherwise, each channel defaults to a certain DGS line as shown in the table. Take care to keep track of the connections to DGS lines. The best way to do this is by specifying the DGS in the channel map.

UltraVI80 Default DGS Lines in a Slices 1 and 2

	DCVI		DCTime			DCVI		DCTime	
Channel	Inst.	DGS	Inst.	DGS	Channel	Inst.	DGS	Inst.	DGS
1	1	1	1	1	21	21	3	1	1
2	2	1	1	1	22	22	3	1	1
3	3	1	2	2	23	23	3	2	2
4	4	1	2	2	24	24	3	2	2
5	5	1	1	1	25	25	3	1	1
6	6	1	1	1	26	26	3	1	1

7	7	1	2	2	27	27	3	2	2
8	8	1	2	2	28	28	3	2	2
9	9	1	1	1	29	29	3	1	1
10	10	1	1	1	30	30	3	1	1
11	11	2	2	2	31	31	4	2	2
12	12	2	2	2	32	32	4	2	2
13	13	2	1	1	33	33	4	1	1
14	14	2	1	1	34	34	4	1	1
15	15	2	2	2	35	35	4	2	2
16	16	2	2	2	36	36	4	2	2
17	17	2	1	1	37	37	4	1	1
18	18	2	1	1	38	38	4	1	1
19	19	2	2	2	39	39	4	2	2
20	20	2	2	2	40	40	4	2	2

UltraVI80 Default DGS Lines in a Slices 3 and 4

	DCVI		DCTime			DCVI		DCTime	
Channel	Inst.	DGS	Inst.	DGS	Channel	Inst.	DGS	Inst.	DGS
41	41	5	3	5	61	61	7	3	5
42	42	5	3	5	62	62	7	3	5
43	43	5	4	6	63	63	7	4	6
44	44	5	4	6	64	64	7	4	6
45	45	5	3	5	65	65	7	3	5
46	46	5	3	5	66	66	7	3	5
47	47	5	4	6	67	67	7	4	6
48	48	5	4	6	68	68	7	4	6
49	49	5	3	5	69	69	7	3	5
50	50	5	3	5	70	70	7	3	5

51	51	6	4	6	71	71	8	4	6
52	52	6	4	6	72	72	8	4	6
53	53	6	3	5	73	73	8	3	5
54	54	6	3	5	74	74	8	3	5
55	55	6	4	6	75	75	8	4	6
56	56	6	4	6	76	76	8	4	6
57	57	6	3	5	77	77	8	3	5
58	58	6	3	5	78	78	8	3	5
59	59	6	4	6	79	79	8	4	6
60	60	6	4	6	80	80	8	4	6

DCDiffMeter DGS

The DCDiffMeter is only connected to a DGS line when specified in a channel map for use on the Low Side of the DCDiffMeter. For more information, see Programming the Channel Map in the DCDiffMeter section.

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UltraVI80 Instrument Board : Using UltraVI80 in Concurrent Testing

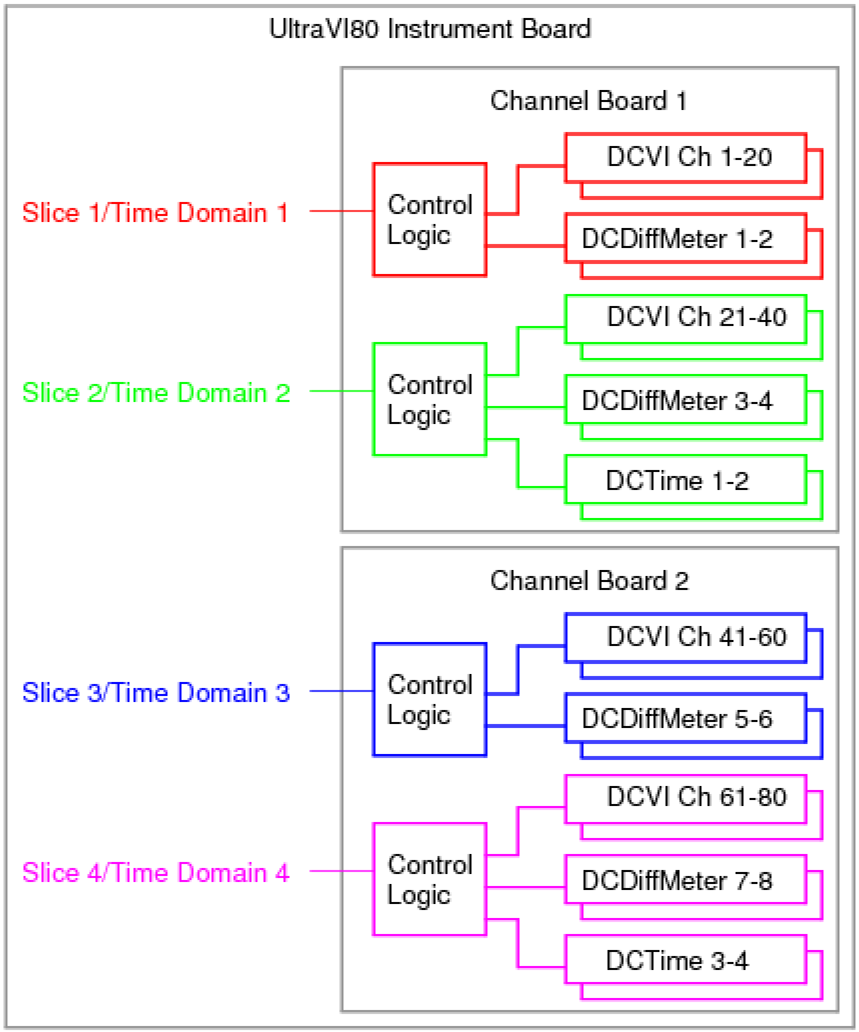
Using UltraVI80 in Concurrent Testing

Each UltraVI80 channel board slice can be assigned to a unique time domain:

- Up to four slices total for each UltraVI80 instrument board
  - Slices 2 and 4 include: 20 DCVI channels, two DC Differential Voltmeters, and two DC Time Measurement Units
  - Slices 1 and 3 include: 20 DCVI channels and two DC Differential Voltmeters
- Each UltraVI80 instrument slice can be controlled by unique time domains simultaneously using Sync-Link. Features include:
  - Independent access to Sync-Link
  - Independent sample clock programming
  - Independent clock synchronization with other UltraVI80 instruments or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers

- Multiple slices can be grouped into a common time domain
- UltraVI80 fully supports flow domains and asynchronous pattern starts

The following diagram shows the concurrent testing architecture for the UltraVI80.



UltraVI80 Concurrent Testing Architecture

For more information, see About Concurrent Test.

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UltraVI80 Instrument Board : UltraVI80 Errors

UltraVI80 Errors

An error is a condition that is determined by the hardware at run time and does not represent a risk of hardware damage. An error occurs when the hardware does enter a programmed state or does not initiate a requested process.

The UltraVI80 Instrument Board can have the following errors:

DCTime Connect	A DCTime Instrument was illegally connected. DCTime instruments can only be connected to one channel (.sense#) or High Performance input (.dig#) at a time.
DCDiffMeter Highside/Lowside	Multiple Highside or Lowside connections were attempted to a DCDiffMeter. Check your patterns and verify that only one DCDiffMeter Connect or Disconnect microcode is being issued per DCDiffMeter in a vector.
DCTime Missed Microcode	The execution of a DCTime microcode or PSet was missed. Check your patterns and adjust the timing to ensure that there is at least 200ns between vectors.
Microcode Download	An error occurred while downloading a pattern.
Site Programming	A site programming error occurred. Check your test program and verify that there is at least 5s between site changes in site loops.
DCSubSystem Over Voltage	An over voltage condition was detected while using the UltraVI80.
VBias Over Temperature	An over temperature condition was detected while using the UltraVI80 VBias capability.

Note that all error messages include the slot # where the error occurred and, where possible, the instrument names or channel numbers affected.

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UltraVI80 Instrument Board : UltraVI80 Shutdowns

UltraVI80 Shutdowns

A shutdown indicates that a condition exists that could potentially damage an instrument. Repeatedly invoking a shutdown on any instrument can adversely affect the reliability of all instruments in the test system.

The UltraVI80 responds to or invokes a system shutdown and then executes a shutdown sequence. This is a sequence of events that a UltraVI80 goes through to protect itself from damage and to remove potentially dangerous conditions. Whenever an instrument shutdown occurs, each instrument that responds to the shutdown also executes a shutdown sequence specific to that instrument.

The following topics are included:

- DCTime Shutdown Conditions
- Clearing a Shutdown Condition

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UltraVI80 Instrument Board : UltraVI80 Shutdowns : DCTime Shutdown Conditions

DCTime Shutdown Conditions

A DCTime invokes a system shutdown when the following condition exists:

- Over Voltage

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UltraVI80 Instrument Board : UltraVI80 Shutdowns : Over Voltage

Over Voltage

The DCTime input voltage has exceeded the programmed front end voltage range.

Shutdown Message

The following are the datalog shutdown messages:

{-1} : alarm ["PinName", Site #] TMU:0015 : POSITIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

{-1} : alarm ["PinName", Site #] TMU:0016 : NEGATIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

Common Causes

An over voltage shutdown can be caused by the following:

- Defective device
- Incorrect front end range selection

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UltraVI80 Instrument Board : UltraVI80 Shutdowns : Clearing a Shutdown Condition

Clearing a Shutdown Condition

To clear a system shutdown, IG-XL must call an initialize routine. This means that:

- The test program in which the shutdown occurred must be stopped.
- DataTool must be stopped and started.
- The job must be revalidated.

Following a shutdown clear, make sure to understand and resolve the cause of the shutdown before rerunning the program.

Shutdown Sequence

For more information, see Shutdowns.

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UltraVI80 Instrument Board : UltraVI80 Software Licensing Requirements

UltraVI80 Software Licensing Requirements

IG-XL enforces counted licenses for each UltraVI80 DCVI channel, and each DCVI channel is counted as one license.

For merged channels, one license is required for the merged pair. The leader channel requires a license, but the follower channel does not.

The license check is done at validation time. Each UltraVI80 DCVI channel in the channel map requires one license. DCTime and DCDiffMeter do not count for licensing. Licenses are checked back in before revalidation or when you quit IG-XL. No licenses are required when running calibration or checkers.

For more information, see IG-XL Licensing.

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UltraVI80 Instrument Board

UltraVI80 Instrument Board

The UltraVI80 DCVI is a multichannel DC instrument board designed to source and sink current and voltage, measure current and voltage, and make differential and time measurements, allowing multiple test applications on a DUT.

The UltraVI80 instrument board provides hardware that supports the following DC instruments:

- 80 DCVI channels
- 8 DC Differential Voltmeters (DCDiffMeters)
- 4 DC Time Measurement Units (DCTimes)

The following topics are available:

- Features
- UltraVI80 Board Hardware
- Test Head Connections
- Simulated Configuration Files
- UltraVI80 Channel Connections

- Connecting to a Live DUT from Force (V or I) Mode
- Device Ground Sense
- Using UltraVI80 in Concurrent Testing
- UltraVI80 Errors
- UltraVI80 Shutdowns
- UltraVI80 Software Licensing Requirements

For more information on UltraVI80 instruments, see:

- About the UltraVI80 DCDiffMeter
- About the UltraVI80 DCTime
- About the UltraVI80 DCVI
- UltraVI80 Debug Displays
- UltraVI80 DIB Design Guidelines

For information on UltraVI80 instrument performance specifications and standards, see UltraVI80 Specifications.

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